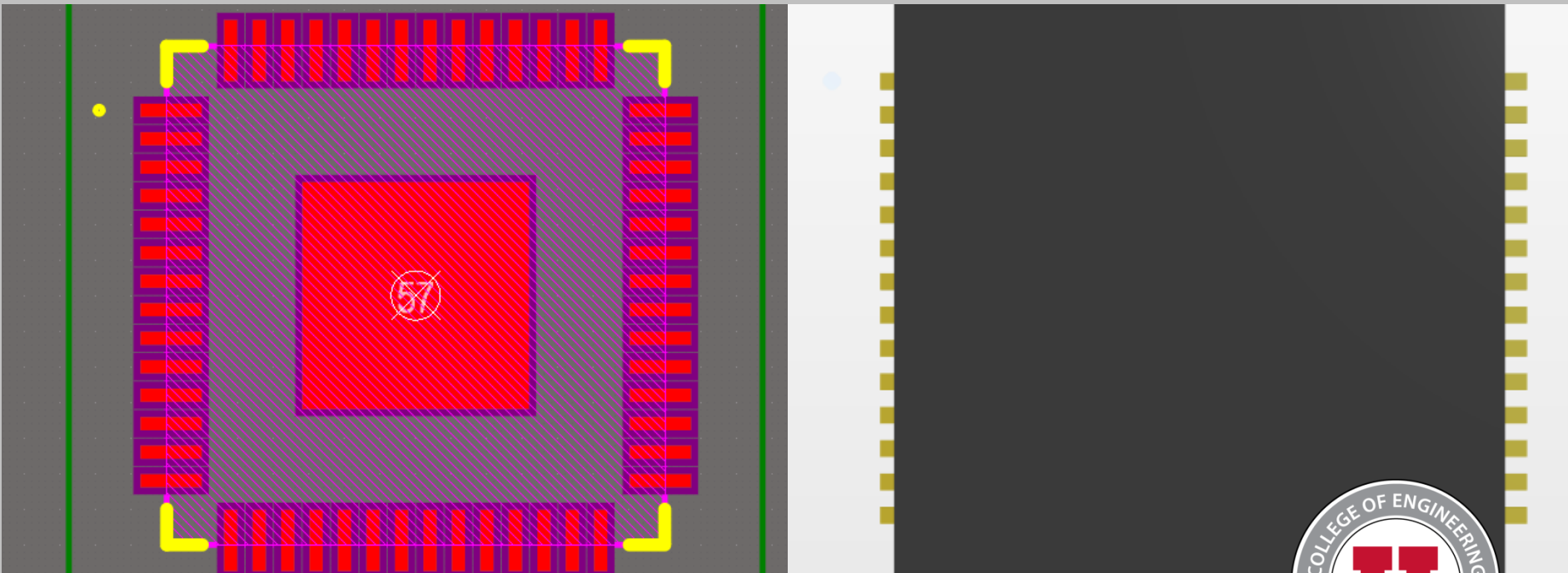


IEEE PCB Design Workshop: (Week 04) Footprints and PCB Layout



Hosted By: Adrian Sucahyo and IEEE at the University of Utah



Adapted From: IEEE x FSAE Workshop SP25 with Nick Howard and Adrian Sucahyo

Workshop Outline

Schedule:

- Feb. 17 – Introduction to Schematics
- Feb. 24 – Datasheets and Components
- Mar. 3 – Introduction to PCB Layout
- ** SPRING BREAK **
- Mar. 17 – Layout Continued
- Mar. 24 – Soldering Week 1
- Mar. 31 – Soldering Week 2
- Apr. 3 – Soldering Week 3

Announcements

- Alternative Projects
 - We will be able to get alternative project boards manufactured if submitted by the deadline.
 - Limited to the 10 cm x 10 cm dimensions outlined by JLCPCB.
 - Talk or email me if you have any questions!
- PCB Submission Deadlines
 - For both workshop and custom PCBs, the deadline for submission is **March 20th**

Want more experience?

- Consider joining the FSAE tractive team!
 - The Tractive Team is currently looking for students to assist with designing and assembling the electrical system for an electric formula-style race car!
 - No experience required!



U of U FSAE Discord Link



Join the IEEE Discord

- If you haven't already, please join the IEEE Discord server for additional information and updates regarding this workshop

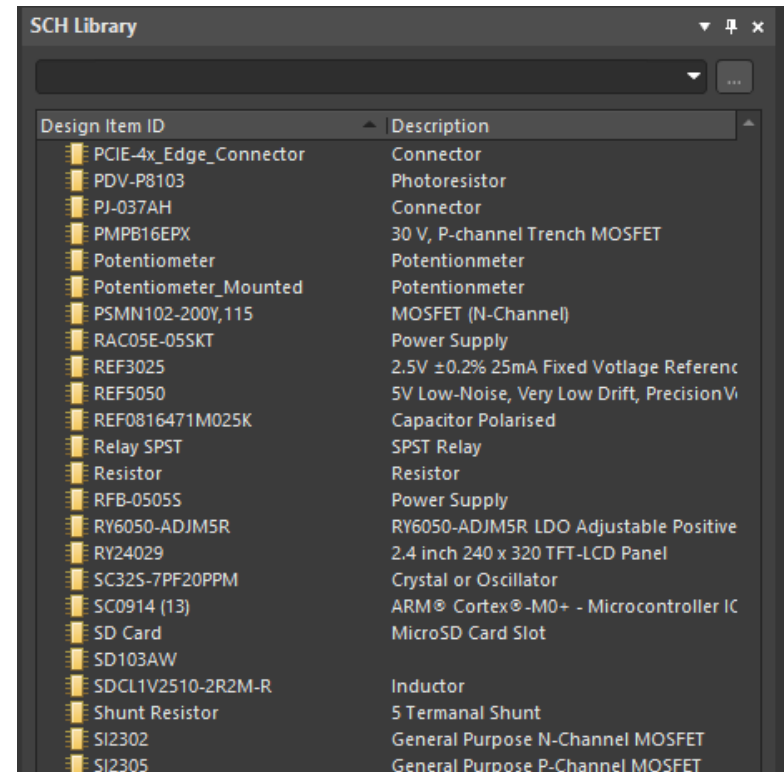


U of U IEEE Discord Link



Component Libraries Refresher

- Libraries are collections of components and footprints
 - All components for a project must be derived from a library
 - Projects may reference multiple different component libraries
 - Relatively consistent across platforms
- Typically, there are symbol and footprint libraries
 - Tightly coupled together

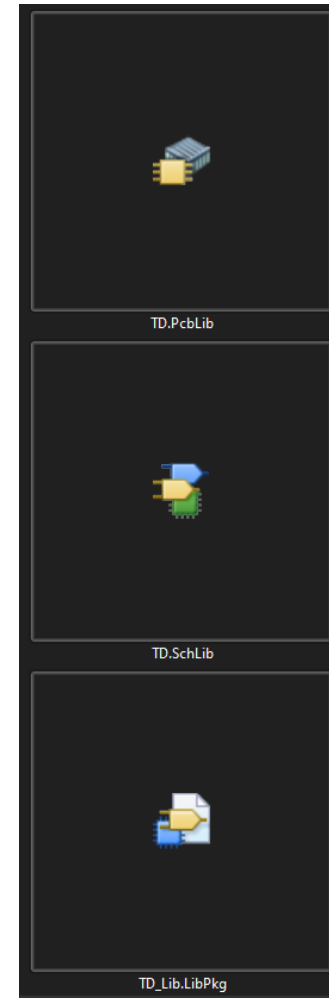


The screenshot shows a software window titled "SCH Library". It contains a table with two columns: "Design Item ID" and "Description". The table lists various electronic components and their specifications.

Design Item ID	Description
PCIE-4x_Edge_Connector	Connector
PDV-P8103	Photoresistor
PJ-037AH	Connector
PMPB16EPX	30 V, P-channel Trench MOSFET
Potentiometer	Potiontometer
Potentiometer_Mounted	Potiontometer
PSMN102-200Y,115	MOSFET (N-Channel)
RAC05E-05SKT	Power Supply
REF3025	2.5V $\pm$ 0.2% 25mA Fixed Voltage Referenc
REF5050	5V Low-Noise, Very Low Drift, Precision V
REF0816471M025K	Capacitor Polarised
Relay SPST	SPST Relay
Resistor	Resistor
RFB-05055	Power Supply
RY6050-ADJM5R	RY6050-ADJM5R LDO Adjustable Positive
RY24029	2.4 inch 240 x 320 TFT-LCD Panel
SC325-7PF20PPM	Crystal or Oscillator
SC0914 (13)	ARM [®] Cortex [®] -M0+ - Microcontroller IC
SD Card	MicroSD Card Slot
SD103AW	
SDCL1V2510-2R2M-R	Inductor
Shunt Resistor	5 Termanal Shunt
SI2302	General Purpose N-Channel MOSFET
SI2305	General Purpose P-Channel MOSFET

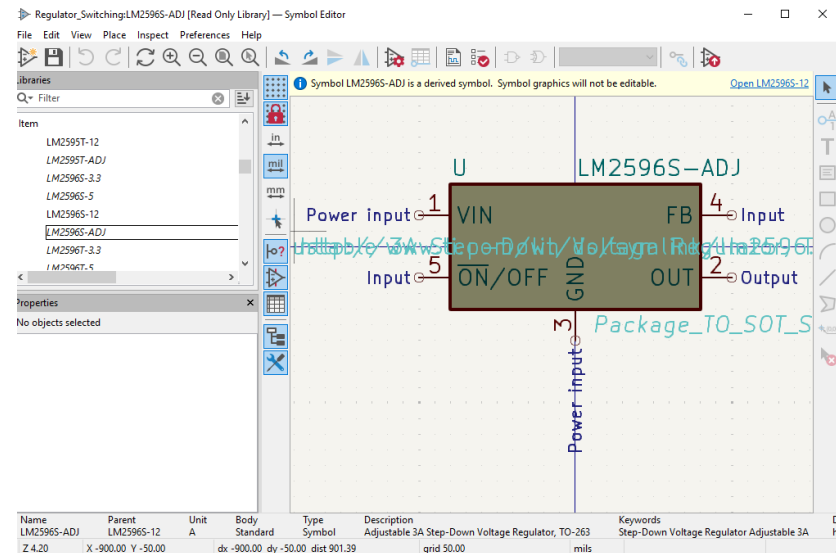
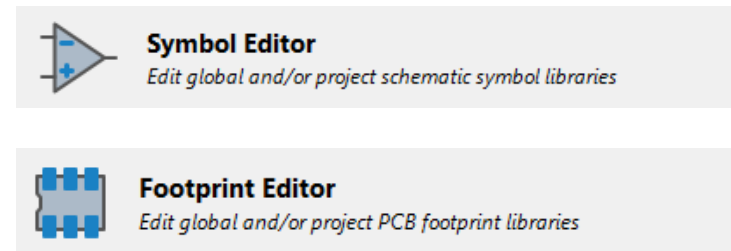
Component Library Types (Altium)

- .SchLib
 - Contains all component schematic symbols
- .PcbLib
 - Contains all footprint information
- .LibPkg
 - Also known as an integrated library
 - Packages multiple .SchLib and .PcbLib libraries together for easier management



Component Library Types (KiCAD)

- .kicad_sym
 - Contains all the symbol information for components in the library
- .pretty
 - Contains all the footprint information to be used with components



Standard vs. Non-Standard Footprints

- Many components are in “standard” packages
 - Key parameters are standardized across manufacturers
- Certain components are in “non-standard” footprints due to custom features or other restrictions



<https://www.indiamart.com/proddetail/smd-electronic-component-2856299588912.html>

Resistors and Capacitors

- Many components are in “standard” packages
 - Key parameters are standardized across manufacturers
- Certain components are in “non-standard” footprints due to custom features or other restrictions

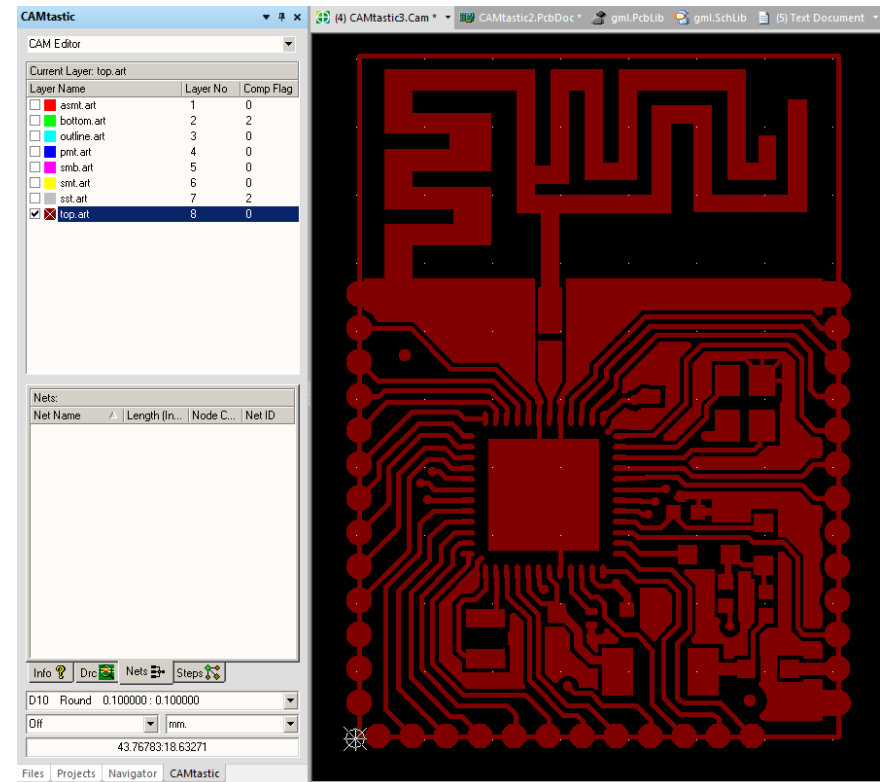
Imperial Size Code		Metric Size Code	
 .1"x.1" (100mils)  .5"x.5" (500mils) Size Reference	01005	.	0402
	0201	.	0603
	0402	-	1005
	0603	-	1608
	0805	-	2012
	1008	-	2520
	1206	-	3216
	1210	-	3225
 1x1cm Size Reference	1806	-	4516
	1812	-	4532
	2010	-	5025
	2512	-	6332

Component Sizes (common in bold)

https://developer.wildernesslabs.co/Hardware/Reference/Components/Packages_and_Sizes/

Gerber Files

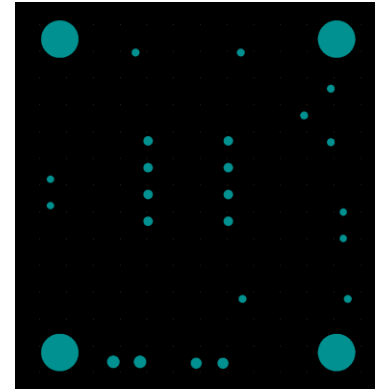
- Gerbers (or Gerber Files) are the files sent to the manufacturer that will be used to make the boards
- Engineers at the fab review files for manufacturability and then send it off to production.
- Gerber files can be generated from EDA software but needs to be reviewed.



<https://www.onethesis.com/gerbers-to-footprint/>

NC Drill Files

- NC Drill Files contain the information for the drill holes of the PCB.
 - These are generated separately from the Gerber Files



----- NCDrill File Report For: Workshop Project.PcbDoc 10/29/2024 1:09:38 AM -----						
Layer Pair : Top Layer to Bottom Layer ASCII RoundHoles File : Workshop Project.TXT						
Tool	Hole Size	Hole Tolerance	Hole Type	Hole Count	Plated	Tool Travel

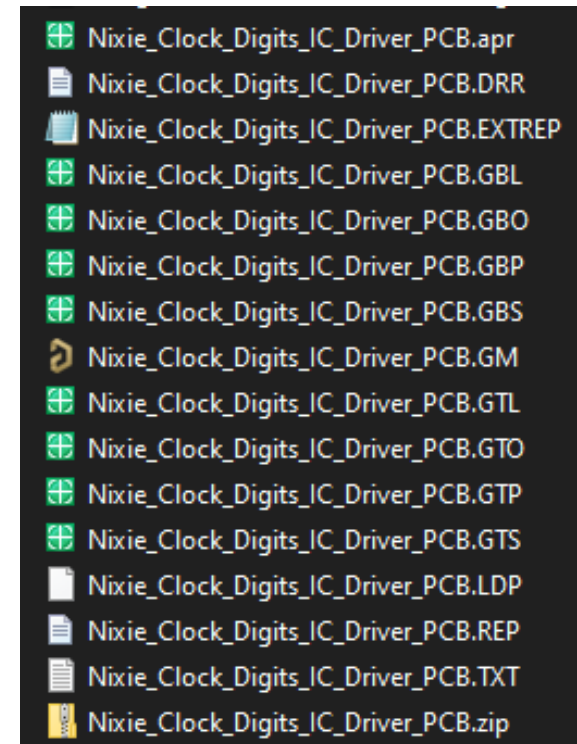
T1	28mil (0.7mm)		Round	4	PTH	1.29inch (32.82mm)
T2	30mil (0.75mm)		Round	7	PTH	2.02inch (51.35mm)
T3	35mil (0.9mm)		Round	8	PTH	0.90inch (22.86mm)
T4	41mil (1.05mm)		Round	2	PTH	0.10inch (2.54mm)
T5	47mil (1.19mm)		Round	2	PTH	0.10inch (2.54mm)
T6	140mil (3.556mm)		Round	4	PTH	3.24inch (82.30mm)

Totals				27		

Total Processing Time (hh:mm:ss) : 00:00:00						

Gerber File and Drill File Types

- Each file represents a different layer or operation type.
- Common files:
 - .gto, .gbo = top/bottom silkscreen
 - .gts, .gbs = top/bottom solder mask
 - .gtp, .gbp = top/bottom solder paste
 - .gtl, .gbl = top/bottom copper
 - .gm = board outline
 - .drl / .txt = CNC drill



Questions?

Questions?

Download Today's Project Files

Navigate to the workshop GitHub and
download today's files

<https://github.com/IEEE-U-of-U/IEEE-PCB-Workshop-Spring-2026>